
Process Improvement

Objectives

- To explain the principles of software process improvement
- To explain how software process factors influence software quality and productivity
- To explain how to develop simple models of software processes
- To explain the notion of process capability and the CMMI process improvement model

Topics covered

- Process and product quality
- Process classification
- Process measurement
- Process analysis and modelling
- Process change
- The CMMI process improvement framework

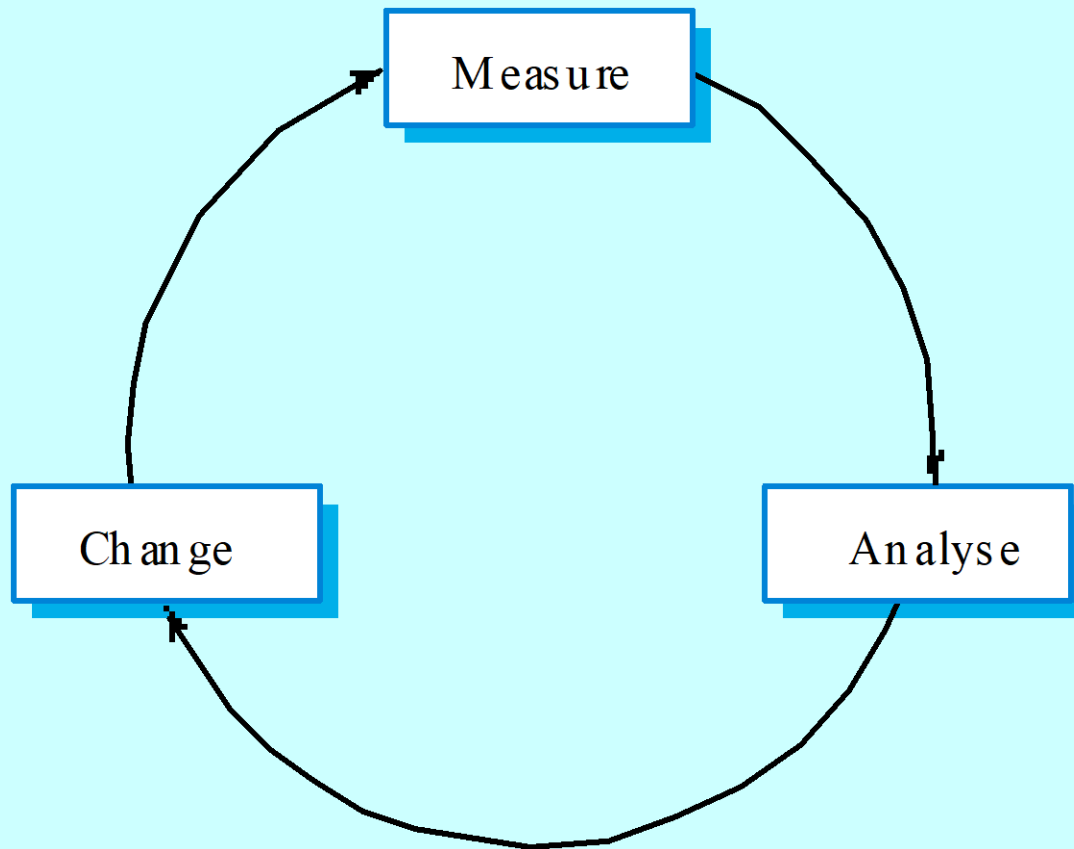
Process improvement

- Understanding existing processes and introducing process changes to improve product quality, reduce costs or accelerate schedules.
- Most process improvement work so far has focused on defect reduction. This reflects the increasing attention paid by industry to quality.
- However, other process attributes can also be the focus of improvement

Process attributes

Process characteristic	Description
Understandability	To what extent is the process explicitly defined and how easy is it to understand the process definition?
Visibility	Do the process activities culminate in clear results so that the progress of the process is externally visible?
Supportability	To what extent can CASE tools be used to support the process activities?
Acceptability	Is the defined process acceptable to and usable by the engineers responsible for producing the software product?
Reliability	Is the process designed in such a way that process errors are avoided or trapped before they result in product errors?
Robustness	Can the process continue in spite of unexpected problems?
Maintainability	Can the process evolve to reflect changing organisational requirements or identified process improvements?
Rapidity	How fast can the process of delivering a system from a given specification be completed?

The process improvement cycle



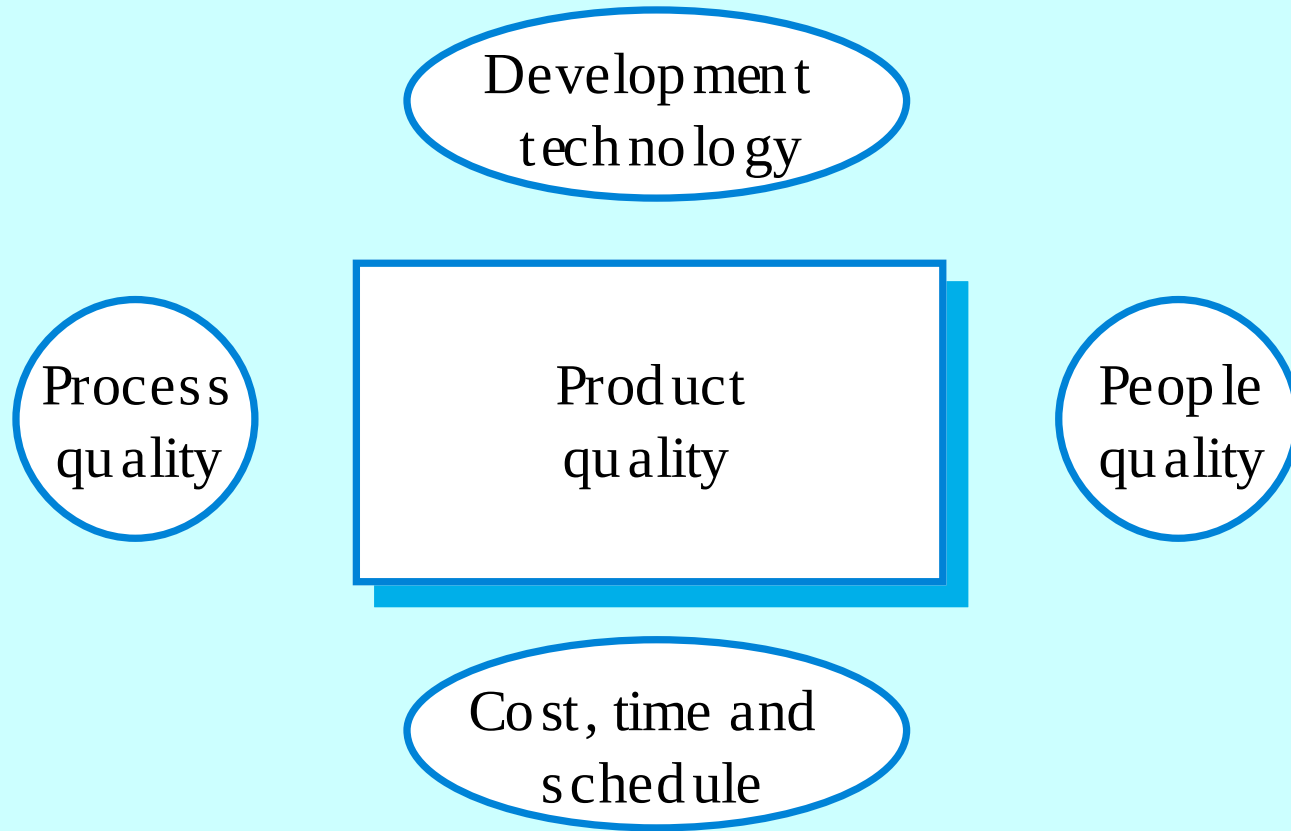
Process improvement stages

- **Process measurement**
 - Attributes of the current process are measured. These are a baseline for assessing improvements.
- **Process analysis**
 - The current process is assessed and bottlenecks and weaknesses are identified.
- **Process change**
 - Changes to the process that have been identified during the analysis are introduced.

Process and product quality

- Process quality and product quality are closely related and process improvement benefits arise because the quality of the product depends on its development process.
- A good process is usually required to produce a good product.
- For manufactured goods, process is the principal quality determinant.
- For design-based activity, other factors are also involved especially the capabilities of the designers.

Principal product quality factors



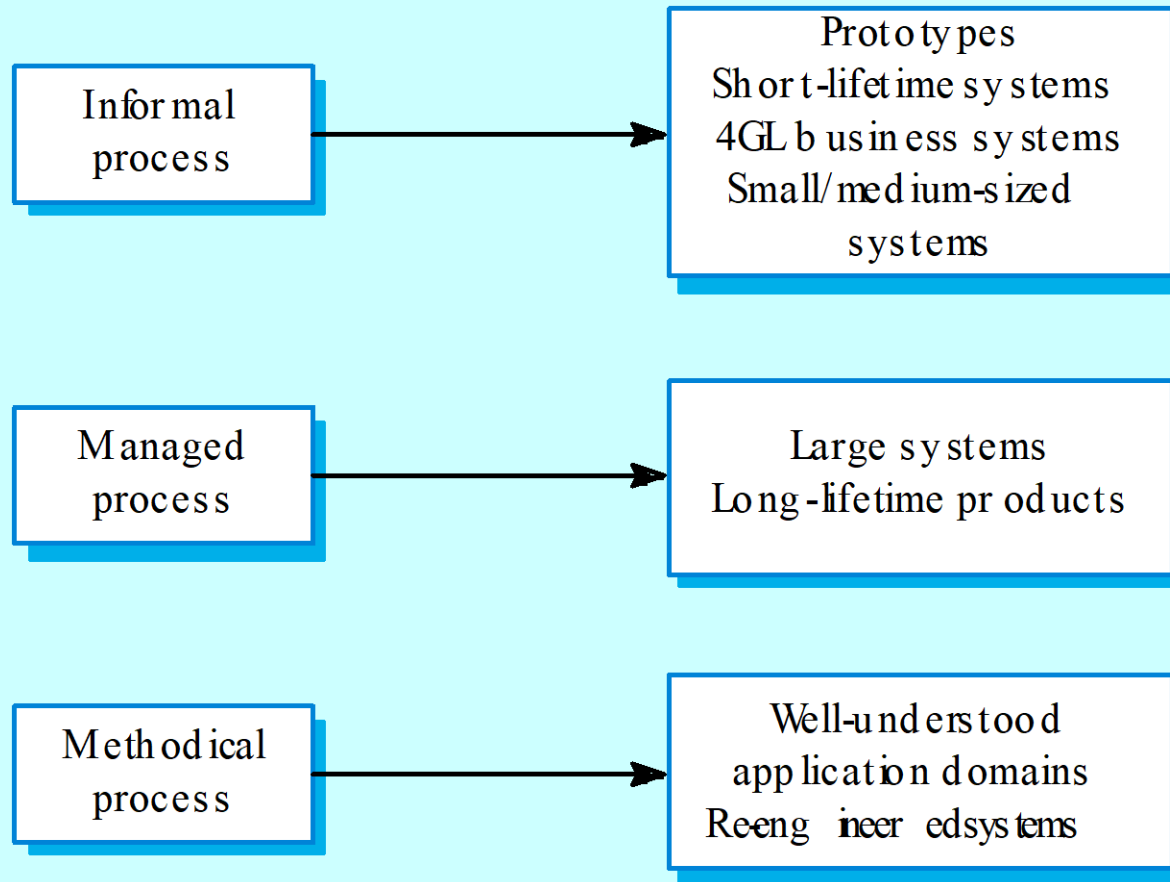
Quality factors

- For large projects with 'average' capabilities, the development process determines product quality.
- For small projects, the capabilities of the developers is the main determinant.
- The development technology is particularly significant for small projects.
- In all cases, if an unrealistic schedule is imposed then product quality will suffer.

Process classification

- Informal
 - No detailed process model. Development team chose their own way of working.
- Managed
 - Defined process model which drives the development process.
- Methodical
 - Processes supported by some development method such as the RUP.
- Supported
 - Processes supported by automated CASE tools.

Process applicability



Process measurement

- Wherever possible, quantitative process data should be collected
 - However, where organisations do not have clearly defined process standards this is very difficult as you don't know what to measure. A process may have to be defined before any measurement is possible.
- Process measurements should be used to assess process improvements
 - But this does not mean that measurements should drive the improvements. The improvement driver should be the organizational objectives.

Classes of process measurement

- Time taken for process activities to be completed
 - E.g. Calendar time or effort to complete an activity or process.
- Resources required for processes or activities
 - E.g. Total effort in person-days.
- Number of occurrences of a particular event
 - E.g. Number of defects discovered.

Goal-Question-Metric Paradigm

- Goals
 - What is the organisation trying to achieve? The objective of process improvement is to satisfy these goals.
- Questions
 - Questions about areas of uncertainty related to the goals. You need process knowledge to derive these.
- Metrics
 - Measurements to be collected to answer the questions.

Process analysis and modelling

- Process analysis
 - The study of existing processes to understand the relationships between parts of the process and to compare them with other processes.
- Process modelling
 - The documentation of a process which records the tasks, the roles and the entities used;
 - Process models may be presented from different perspectives.

Process analysis and modelling

- Study an existing process to understand its activities.
- Produce an abstract model of the process. You should normally represent this graphically. Several different views (e.g. activities, deliverables, etc.) may be required.
- Analyse the model to discover process problems. This involves discussing process activities with stakeholders and discovering problems and possible process changes.

Process analysis techniques

- Published process models and process standards
 - It is always best to start process analysis with an existing model. People then may extend and change this.
- Questionnaires and interviews
 - Must be carefully designed. Participants may tell you what they think you want to hear.
- Ethnographic analysis
 - Involves assimilating process knowledge by observation. Best for in-depth analysis of process fragments rather than for whole-process understanding.

The CMMI framework

- The CMMI framework is the current stage of work on process assessment and improvement that started at the Software Engineering Institute in the 1980s.
- The SEI's mission is to promote software technology transfer particularly to US defence contractors.
- It has had a profound influence on process improvement
 - Capability Maturity Model introduced in the early 1990s.
 - Revised maturity framework (CMMI) introduced in 2001.

Process capability assessment

- Intended as a means to assess the extent to which an organisation's processes follow best practice.
- My providing a means for assessment, it is possible to identify areas of weakness for process improvement.
- There have been various process assessment and improvement models but the SEI work has been most influential.

The SEI capability maturity model

- Initial
 - Essentially uncontrolled
- Repeatable
 - Product management procedures defined and used
- Defined
 - Process management procedures and strategies defined and used
- Managed
 - Quality management strategies defined and used
- Optimising
 - Process improvement strategies defined and used

Problems with the CMM

- Practices associated with model levels
 - Companies could be using practices from different levels at the same time but if all practices from a lower level were not used, it was not possible to move beyond that level
- Discrete rather than continuous
 - Did not recognise distinctions between the top and the bottom of levels
- Practice-oriented
 - Concerned with how things were done (the practices) rather than the goals to be achieved.

The CMMI model

- An integrated capability model that includes software and systems engineering capability assessment.
- The model has two instantiations
 - Staged where the model is expressed in terms of capability levels;
 - Continuous where a capability rating is computed.

CMMI model components

- **Process areas**
 - 24 process areas that are relevant to process capability and improvement are identified. These are organised into 4 groups.
- **Goals**
 - Goals are descriptions of desirable organisational states. Each process area has associated goals.
- **Practices**
 - Practices are ways of achieving a goal - however, they are advisory and other approaches to achieve the goal may be used.

CMMI process areas 1

Process management

- Organisational process definition
- Organisational process focus
- Organisational training
- Organisational process performance
- Organisational innovation and deployment

Project management

- Project planning
- Project monitoring and control
- Supplier agreement management
- Integrated project management
- Risk management
- Integrated teaming
- Quantitative project management

CMMI process areas 2

Engineering

- Requirements management
- Requirements development
- Technical solution
- Product integration
- Verification
- Validation

Support

- Configuration management
- Process and product quality management
- Measurement and analysis
- Decision analysis and resolution
- Organisational environment for integration
- Causal analysis and resolution

CMMI goals

Goal	Process area
Corrective actions are managed to closure when the project's performance or results deviate significantly from the plan.	Specific goal in Project Monitoring and Control
Actual performance and progress of the project is monitored against the project plan.	Specific goal in project monitoring and control
The requirements are analysed and validated and a definition of the required functionality is developed.	Specific goal in requirements development.
Root causes of defects and other problems are systematically determined.	Specific goal in causal analysis and resolution.
The process is institutionalised as a defined process.	Generic goal

CMMI practices

Practice

Analyse derived requirements to ensure that they are necessary and sufficient

Validate requirements to ensure that the resulting product will perform as intended in the user's environment using multiple techniques as appropriate.

Select the defects and other problems for analysis.

Perform causal analysis of selected defects and other problems and propose actions to address them.

Establish and maintain an organisational policy for planning and performing the requirements development process.

Assign responsibility and authority for performing the process, developing the work products and providing the services of the requirements development process.

Associated goal

The requirements are analysed and validated and a definition of the required functionality is developed.

Root causes of defects and other problems are systematically determined.

The process is institutionalised as a defined process.

CMMI assessment

- Examines the processes used in an organisation and assesses their maturity in each process area.
- Based on a 6-point scale:
 - Not performed;
 - Performed;
 - Managed;
 - Defined;
 - Quantitatively managed;
 - Optimizing.